RATIONAL POINTS ON THE NON-SPLIT CARTAN MODULAR CURVE OF LEVEL 121

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Abstract. We determine the rational points on the non-split Cartan modular curve $X_{\mathrm{ns}}^{+}(121)$. It has exactly seven rational points, all of which are CM. This completes the classification of the 11-adic Galois images of non-CM elliptic curves over $\mathbb{Q}$. A cusp descent reduces the problem to seven curves of genus five over a number field. We determine their relevant points by an isogeny descent and Selmer-group Chabauty. The proof uses a local formal-division calculation at points with singular reduction and does not assume the generalized Riemann hypothesis.

1. Introduction

Let $C_{\mathrm{ns}}(121)$ be a non-split Cartan subgroup of $\mathrm{GL}_2(\mathbb{Z}/121\mathbb{Z})$, obtained from an unramified quadratic algebra, and let $N_{\mathrm{ns}}(121)$ be its normalizer. We write $X_{\mathrm{ns}}^{+}(121)$ for the corresponding modular curve. It has genus 511 and parametrizes elliptic curves with a non-split Cartan structure of level 121, modulo the normalizer.

Rouse, Sutherland and Zureick-Brown classified the possible 11-adic Galois images of non-CM elliptic curves over $\mathbb{Q}$, apart from the images contained in the inverse image of $N_{\mathrm{ns}}(121)$ [4, Theorem 1.6]. The remaining case is equivalent to determining whether $X_{\mathrm{ns}}^{+}(121)$ has a non-CM rational point. The natural map

$$X_{\mathrm{ns}}^{+}(121) \longrightarrow X_{\mathrm{ns}}^{+}(11)$$

has degree 121. The target is an elliptic curve of rank one. Schoof and Tzanakis determined its rational points with integral j-invariant [5]. However, that result does not exclude points with nonintegral j-invariant coming from level 121.

In this paper we determine all rational points on $X_{\mathrm{ns}}^{+}(121)$. The main result is the following theorem.

Theorem 1.1. *The curve $X_{\mathrm{ns}}^{+}(121)$ has exactly seven rational points. They are CM points with discriminants*

$$-3,\ -4,\ -12,\ -16,\ -27,\ -67,\ -163.$$

There is exactly one point for each discriminant. There are no rational cusps and no non-CM rational points.

For an elliptic curve $A/\mathbb{Q}$, write $\rho_{A,11^\infty} : G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{Z}_{11})$ for the action on its Tate module and $\rho_{A,121}$ for its reduction modulo 121. We obtain the following consequence.

Corollary 1.2. *Let $A/\mathbb{Q}$ be a non-CM elliptic curve. Then $\rho_{A,121}(G_{\mathbb{Q}})$ is not contained in a conjugate of $N_{\mathrm{ns}}(121)$. In particular, the classification of 11-adic Galois images of non-CM elliptic curves over $\mathbb{Q}$ is complete.*

Proof. Such a containment would give a non-CM rational point on $X_{\mathrm{ns}}^{+}(121)$, contrary to Theorem 1.1. The last assertion follows from [4, Theorem 1.6]. □

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We prove Theorem 1.1 by working with the image on $X_{\mathrm{ns}}^{+}(11)$. A cusp descent and a sieve at six primes leave seven possibilities for the descent classes, together with a congruence modulo 4356. Each class is represented by a known CM point. A further two-descent gives seven auxiliary curves of genus five over a number field of degree 15. We retain the condition that their points arise from rational points on $X_{\mathrm{ns}}^{+}(11)$.

We then apply the isogeny version of Selmer-group Chabauty [6, Theorem 2.1 and Remark 2.2]. We compute the relevant Selmer groups after adjoining ζ_{11} and compare their localizations at 11 with images obtained by repeated formal division. While a point remains sufficiently deep in the formal group, injectivity of Selmer localization forces a global division. The final local image is incompatible with the computed Selmer image, which excludes an additional rational point.

The local calculation is different in the odd and even classes. In the even classes the known point has smooth reduction. In the odd classes it reduces to the singular point of the integral model, and we work over ramified quadratic completions. We give the formal-group calculation in both cases. The descent requires neither a basis of the Mordell-Weil groups of the auxiliary Jacobians nor a height bound for the image on $X_{\mathrm{ns}}^{+}(11)$.

The number-field computations use an unconditional discriminant bound to certify the required class-group input. GRH is used only to find candidate generators, which are then certified by exact membership and local independence. Thus Theorem 1.1 does not depend on GRH.

1.1. **Notation.** For a number field k, a finite set S of finite primes and a prime p, put

$$k(S,p) = \{a \in k^{\times}/k^{\times p} : v_{\mathfrak{q}}(a) \equiv 0 \pmod{p} \text{ for } \mathfrak{q} \notin S\}.$$

We use additive notation for this $\mathbb{F}_p$-vector space. Valuations on local fields are normalized to have value group $\mathbb{Z}$. If λ is an isogeny, μ_v denotes its local Kummer map.

1.2. **The CM points.** For each discriminant in Theorem 1.1, the CM order has class number one and 11 is inert. Its action on the torsion gives a rational point on $X_{\mathrm{ns}}^{+}(121)$. The corresponding j-invariants and the images on $X_{\mathrm{ns}}^{+}(11)$ are listed in Table 1.

We shall use uniqueness of the rational CM lift from level 11. Lifts of a fixed Cartan algebra modulo 11 form an affine space whose translation space has dimension two over $\mathbb{F}_{11}$. The canonical CM lift is the origin. The Cartan acts on the translation space through the norm-one subgroup of $\mathbb{F}_{121}^{\times}$, which has order 12 and acts freely away from the origin. Modulo elliptic automorphisms, the nonzero orbit lengths are 12, 6 or 4. CM reciprocity gives this action, so only the canonical lift is rational. The finite orbit calculation is included in the ancillary data.

2. Cusp descent and the auxiliary curves

The natural map $X_{\mathrm{ns}}^{+}(121) \to X_{\mathrm{ns}}^{+}(11)$ has degree 121. We use the model

$$E = X_{\mathrm{ns}}^{+}(11): \quad y^2 + y = x^3 - x^2 - 7x + 10, \qquad E(\mathbb{Q}) = \mathbb{Z}G, \quad G = (4,5). \tag{2.1}$$

The model and Mordell–Weil generator agree with [5]. The seven known points at level 121 map to the following points.

2.1. **The finite sieve.** Put $K = \mathbb{Q}(z)$, where

$$z^5 + z^4 - 4z^3 - 3z^2 + 3z + 1 = 0, \qquad u = 2 - z.$$

The unique prime above 11 is generated by u. The five cusps are $r_i s$, with

$$(r_i) = (1, 4, 3, 9, 5), \qquad s = (-z^4 - z^3 + 2z^2 + 3z + 3, -2z^4 - 4z^3 + 6z^2 + 11z + 1).$$

The point s has order 11. The cusp divisor is a single closed point of degree five, so E has no rational cusps.

n	nG	D	j
-2	$(2,-1)$	-27	-12288000
-1	$(4,-6)$	-16	287496
0	O	-12	54000
1	$(4,5)$	-67	-147197952000
2	$(2,0)$	-4	1728
3	$(5/4,7/8)$	-3	0
4	$(-2,3)$	-163	-262537412640768000

TABLE 1. The images of the rational points on $X_{\mathrm{ns}}^{+}(121)$.

Suppose a rational point at level 121 maps to $P = nG$. At any prime away from 11 at which $j(P)$ is nonintegral, its valuation is divisible by 121. Indeed, after a quadratic twist the associated elliptic curve is a Tate curve. A nontrivial unipotent inertia element modulo 121 cannot belong to the normalizer of an unramified Cartan: this holds modulo 11, and an element $1 + 11N$, with N nonzero nilpotent modulo 11, cannot belong to the Cartan congruence kernel either. Hence the Tate parameter has valuation divisible by 121. As the polar multiplicity of j at each cusp of E is 11, the intersection multiplicities of P with the cusps are divisible by 11 away from 11.

For $m = 1, 2, 3$, let $a_{m,i}$ be the representative in $\{1, \dots, 10\}$ of $r_i^m \bmod 11$, and choose f_m with

$$\operatorname{div}(f_m) = \sum_i a_{m,i}(r_i s) - 22(O), \qquad f_m(x/y)^{22}\big|_O = 1.$$

The class of $f_m(P)$ lies in $K(S, 11)$, where $S = \{(u)\}$. The class number of K is one. A basis for this five-dimensional space is

$$u, \quad z^4 - 3z^2 + 1, \quad z^2 + z - 1, \quad z, \quad -z^4 - z^3 + 3z^2 + 3z.$$

At O one uses the normalized leading coefficient.

The sieve starts with all three classes unknown. Local residue-symbol equations, with cusp residues retained conservatively, give:

prime	order of $\bar{G}$	modulus	states	ranks of each class system
353	11	11	7	3
23	33	33	10	3 or 5
3301	33	33	7	5
12101	1089	1089	12	5
15731	726	2178	7	5
53	12	4356	7	5

Write $M = 4356 = 4 \cdot 9 \cdot 121$. The final states are exactly

$$n \equiv n_0 \pmod{4356}, \qquad -2 \le n_0 \le 4, \qquad [f_m(nG)] = [f_m(n_0 G)] \quad (m = 1, 2, 3). \tag{2.2}$$

We compute all six local tables from the functions above. The ancillary program `six_prime_sieve.py` performs the exhaustive affine linear algebra and congruence lifting. We compute the classes of the seven known points exactly in the displayed unit basis and verify that they are the surviving classes.

Set

$$\alpha = x(2s) = z^4 - z^3 - 3z^2 + 3z + 2, \qquad \delta_{n_0} = x(n_0 G - 3s) - \alpha.$$

There is an exact normalized function identity

$$\frac{x(P - 3s) - \alpha}{x(3s) - \alpha} = f_1(P)^7 f_2(P)^{10} f_3(P)^6 f_T(P)^3 h(P)^{11},$$

where $h \in K(E)^\times$ and

$$\operatorname{div}(f_T) = 11(s) - 11(O), \qquad f_T(x/y)^{11}\big|_O = 1.$$

We verify the identity, including its constant, in the function field of E. The class of $f_T(P)$ is an unramified descent homomorphism on $E(K)/11E(K)$. It follows from (2.2) that

$$x(nG - 3s) = \alpha + \delta_{n_0} w^{11}, \qquad w \in K. \tag{2.3}$$

Let

$$L = K(\theta), \quad \theta^3 - \theta^2 - 7\theta + 41/4 = 0, \qquad \beta^2 = \alpha - \theta, \quad \beta \in L, \qquad F = L(\zeta_{11}).$$

The degrees of L, F are 15 and 30. The two-descent map $x - \theta$ on $E(L)$ shows that (2.3) gives a point on

$$\eta v^2 = \alpha - \theta + \delta_{n_0} w^{11}, \qquad \eta = \begin{cases} 1 & n_0 \text{ even}, \\ 4 - \theta & n_0 \text{ odd}. \end{cases} \tag{2.4}$$

Indeed, $-3s$ is twice a point and $n \equiv n_0 \bmod 2$. Put $a = \delta_{n_0}/\beta^2$ and $b = a^5$. The ideal of a in F is the square of its unique prime over 2. In particular it is a unit at 11. Over $B = F(\sqrt{\eta})$, the substitutions

$$X = aw, \qquad Y = \sqrt{\eta}\, a^5 v/\beta$$

give the unit model

$$C: \quad Y^2 = X^{11} + b^2. \tag{2.5}$$

Its known point P_0 has $X = a$. We write J for its Jacobian and J^η for its quadratic twist. In an even case $\eta = 1$. We choose the known square root $v_0 \in L$ compatibly with the original elliptic point, and write P_0 for the resulting point with $w = 1$.

2.2. **The odd Selmer groups.** We first consider $n_0 = -1, 1, 3$ and $\eta = 4 - \theta$. Let τ generate $\operatorname{Gal}(B/F)$. The twist identifies $J^\eta(F)$ with the anti-invariant subgroup of $J(B)$. The endomorphisms

$$\sigma(X, Y) = (\zeta_{11} X, Y), \qquad \lambda = 1 - \sigma$$

commute with twisting. The kernel of λ on J is generated by $T = [(0, b) - \infty]$, which has order 11. The action of τ on T is trivial, so $J^\eta[\lambda](F) = 0$. The Kummer map is represented over B by $Y - b$. Its values lie in the anti-invariant part of $B^\times/B^{\times 11}$.

The exact discriminants are

$$|\operatorname{disc} F| = 2^{20} 11^{27}, \qquad |\operatorname{disc} B| = 2^{60} 11^{56}.$$

Both fields have root discriminant less than 19. The row $b = 9$ of Odlyzko's unconditional Table 4 [2] gives

$$|D_M| > 53.047^{r_1} 20.710^{2r_2} e^{-24.001}.$$

For any number field of degree $d \geq 330$, this implies

$$|D_M|^{1/d} > 20.710\, e^{-24.001/330} > 20.710 \left(1 - \frac{24.001}{330}\right) > 19.$$

An everywhere unramified cyclic extension of degree 11 of F or B would have degree at least 330 and the same root discriminant. Thus

$$\operatorname{Cl}(F)[11] = \operatorname{Cl}(B)[11] = 0. \tag{2.6}$$

We verify the discriminants and the displayed rational inequality exactly. Odlyzko rounds the lower bounds down and the error constant up, as explained in [3].

Let S consist of the primes over 2 and 11. Each of F and B has four such primes. Their unit ranks are 14 and 29, and both contain μ_{11}. It follows from (2.6) that

$$\dim F(S, 11) = 19, \qquad \dim B(S, 11) = 34.$$

Norm is onto, since norm composed with restriction is multiplication by 2. Thus the anti-invariant global Kummer space has dimension 15.

We construct this space using candidate generators from the degree-30 fields

$$L(\sqrt{4-\theta}), \qquad L(\sqrt{-11(4-\theta)}).$$

For each candidate we check the ideal valuations outside S, map it into B and divide it by its conjugate. The resulting element has norm one. We obtain fifteen independent joint local images. Since the global space has dimension 15, these classes form a basis. GRH is used to find candidates, but the membership and independence checks certify the basis without that assumption.

There are three primes $v \mid 11$ of F, all with $F_v \simeq \mathbb{Q}_{11}(\zeta_{11})$. At two of them, denoted v_1, v_2, B/F is ramified quadratic. At the third it is unramified quadratic. It is nonsplit at all three, so $J^\eta[\lambda](F_v) = 0$. The determinant of λ on the tangent space has valuation five in F_v, and hence

$$\dim_{\mathbb{F}_{11}} J^\eta(F_v)/\lambda J^\eta(F_v) = 5.$$

We generate each local image using anti-invariant divisors $6(P - \tau P)$, with $P \in C(B_v)$, until its dimension is five. Divisors of degree two are needed to obtain the full local image.

Let $\mathcal{S}^+$ be the intersection of the certified global 15-dimensional space with these three local images. Omitting the condition at 2 makes this a Selmer upper bound, which suffices. We compute that

$$\dim \mathcal{S}^+ = 1 \quad (n_0 = -1, 1, 3), \qquad \mathcal{S}^+ \longrightarrow \prod_{i=1}^{2} B_{v_i}^\times / B_{v_i}^{\times 11} \text{ is injective.} \tag{2.7}$$

The known point has a nonzero class in each of these lines. Thus the actual Selmer group and $J^\eta(F)/\lambda J^\eta(F)$ both have dimension one. Since the rational λ-torsion is zero and λ has cyclotomic norm 11, this forces rank ten over F. The nontrivial automorphism of F/L conjugates the cyclotomic action, so its fixed subspace has half this dimension. Hence the corresponding Jacobian has rank exactly five over L. The rational-point argument below does not use this rank assertion.

3. Local division

We first describe the formal group of the unit model (2.5). At a place v over 11, put $k = F_v$ and let l be either k in an even case or a ramified quadratic completion B_v in an odd case. Put $\pi = 1 - \zeta_{11}$ and $e = v_l(\pi) \in \{1, 2\}$. Thus $v_l(11) = 10e$. For $e = 1$ use $\varpi = \pi$. For $e = 2$ we use

$$\varpi = \frac{\sqrt{\eta}}{\pi^2}, \qquad v_l(\varpi) = 1, \qquad \tau\varpi = -\varpi.$$

In either case the residue field is $\mathbb{F}_{11}$.

We use the proper hyperelliptic model of (2.5). Its special fibre is geometrically integral, with a single cusp and a smooth point at infinity. The formal completion at the identity of its relative Picard group is a smooth formal group of dimension five, since the obstruction group $H^2(\mathcal{O})$ vanishes on each fibre. We may construct its formal coordinates from sums of Abel maps at five smooth sections with distinct residual X-coordinates. The cotangent lattice has basis

$$\omega_j = \frac{X^{j-1}\,dX}{2Y}, \qquad 1 \le j \le 5.$$

Write H_m for formal points whose coordinates lie in $\varpi^m \mathcal{O}_l$.

Lemma 3.1. *The formal logarithm and exponential have integral coefficients in these integral coordinates. Moreover,*

$$\lambda : H_m \xrightarrow{\sim} H_{m+e} \qquad (m \ge 1).$$

In logarithm coordinates dual to the ω_j, the action of λ is diagonal with entries $1 - \zeta_{11}^j$.

Proof. The automorphism σ is the identity on the model modulo the ideal (π), including its nonreduced base change to $\mathcal{O}_l/(\pi)$. Hence every coefficient of the formal power series for $1-\sigma$ is divisible by π. Its linear part divided by π is invertible, since $(1-\zeta_{11}^j)/\pi \equiv j \pmod{\varpi}$. Dividing the series by π and applying the formal inverse function theorem proves the asserted isomorphism.

The norm correspondence $1+\sigma+\cdots+\sigma^{10}$ is zero on the Jacobian, since the quotient curve has genus zero. In $\mathbb{Z}[\sigma]$, multiplication by 11 is therefore a unit times λ^{10}. Every coefficient of its formal series is divisible by π^{10}, and hence by 11. The logarithm identity $\log([11]t) = 11\log(t)$ now proves integrality by induction on total degree. In degree $d \geq 2$, the unknown coefficient is multiplied by $11^d - 11$, whose quotient by 11 is a unit. The already known terms are all divisible by 11. The inverse series is integral as well. Finally $\sigma^*\omega_j = \zeta_{11}^j \omega_j$, which gives the diagonal action. □

3.1. **An odd residue disc.** In this subsection $e = 2$, so $v_l(11) = 20$. Write $f(X) = X^{11} + b^2$.

Lemma 3.2. *Suppose $P_0 = (a, Y_0)$, $v_l(Y_0) = 1$, and $\bar{a} = -1$. If $P = (a+t, Y(t))$ is on the branch through P_0 and $d = v_l(t) \geq 28$, then*

$$[P - P_0] \in H_{d-1} \setminus H_d, \qquad \log[P - P_0] = \frac{t}{2Y_0}(1, a, a^2, a^3, a^4) \pmod{\varpi^d}.$$

Proof. Expand the five differentials in the parameter t, on the analytic branch through P_0. The coefficient of t^h has valuation at least $-2h-1$: use $Y(t)^{-1} = Y_0^{-1}(1 + (f(a+t) - f(a))/Y_0^2)^{-1/2}$, whose binomial coefficients are 11-integral. Upon integration, a term with $h \geq 1$ has valuation at least

$$(h+1)d - 2h - 1 - 20v_{11}(h+1) > d - 1.$$

Thus the five integrals lie in $\varpi^{d-1}\mathcal{O}_l$, and their leading terms are as asserted. Lemma 3.1 allows their exponential in the formal Picard group. This analytic map and the Abel map have identical differentials and agree at P_0. Hence they agree on the disc. The first leading coordinate is nonzero, proving the exact depth. Note that P_0 need not reduce to a smooth point of the model. □

Write H_m^- for the anti-invariant subgroup under τ. For these subgroups one further has

$$H_2^- = H_3^-, \qquad \mu_v(H_2^-) = 0. \tag{3.1}$$

Indeed, τ acts trivially on H_2/H_3, because it fixes the residue field and sends ϖ to $-\varpi$. On an anti-invariant point it acts by -1. Since 2 is invertible, such a point is in H_3. The unique formal λ-division in Lemma 3.1 is again anti-invariant. In particular, the local Kummer class of a point in H_1^- depends only on its leading tangent vector in H_1/H_2.

3.2. **The odd rational-image locus.** Suppose $n \neq n_0$, with $n \equiv n_0 \bmod 4356$, and write $k_0 = v_{11}(n - n_0) \geq 2$. Over the completion of K, normalize $v_K(u) = 1$. The integral minimal elliptic model is

$$x' = (x-4)/u^2, \quad y' = (y+1/2)/u^3, \qquad (y')^2 = (x')^3 + \frac{11}{u^2}(x')^2 + \frac{33}{u^4}x' + \frac{121}{4u^6}.$$

Its discriminant has valuation three. The formal parameter of $2G$ is $-x'/y' = 4u$. The elliptic formal logarithm is an isometry on this formal neighbourhood, since $1 > v_K(11)/10 = 1/2$. Thus

$$v_K(t'((n-n_0)G)) = 1 + 5k_0.$$

For each odd n_0, put $q_0 = n_0 G - 3s$. Exact calculations give

$$v_K(x'(q_0)) = v_K(y'(q_0)) = 1, \qquad v_K(\delta_{n_0}) = 2.$$

The first term in the formal translation of x' is a unit times $2y'(q_0)t'$. Higher terms have larger valuation. Thus

$$v_K(x(nG-3s)-x(q_0))=4+5k_0,$$

and (2.3) gives

$$v_K(w-1)=5k_0-3,\qquad v_{F_v}(X-a)=10k_0-6,\qquad v_l(X-a)=20k_0-12. \tag{3.2}$$

Here $\bar{w}=1$, and the term $11(w-1)$ uniquely determines the valuation of $w^{11}-1$. One also checks $\bar{a}=-1$ and $v_l(Y_0)=1$ at the ramified places. These are the inputs to Lemma 3.2.

It remains to select a single global square root whose signs agree with P_0 at both places. Let

$$R=((n-n_0)/2)G,\qquad Q=2R,\qquad h_\theta(R)=\frac{(x(R)-\theta)^2-(3\theta^2-2\theta-7)}{2y(R)+1}.$$

Then $h_\theta(R)^2=x(Q)-\theta$. If $m=(y(Q)-y(q_0))/(x(Q)-x(q_0))$, set

$$\rho(R)=-\frac{m(\theta-x(q_0))+y(q_0)+1/2}{(x(q_0)-\theta)h_\theta(R)}.$$

The exact function-field identities are

$$\rho(R)^2=\frac{x(q_0+2R)-\theta}{x(q_0)-\theta},\qquad \rho(O)=1.$$

Choose $v=v_0\rho(R)$, where v_0 is the known square root in (2.4). This gives the required global choice. Since $4\mid n-n_0$, the point R is formal. Its original elliptic parameter has F_v-valuation at least 20, while $v_{F_v}(x(q_0)-\theta)\le 6$. The expansion of ρ at O therefore has $\rho(R)-1$ of positive valuation (in fact at least 14). It chooses the branch through P_0 simultaneously. We verify the square identity in the elliptic function field. The value at O and all the displayed valuations are checked exactly.

Let $D=[P-P_0]\in J^\eta(F)$. By Lemma 3.2, its depth at each ramified place is $20k_0-13$. After

$$r=10k_0-7\equiv 3 \pmod{10} \tag{3.3}$$

formal divisions by λ, its depth is one. Up to a nonzero scalar at each place, its leading tangent vector is

$$\mathbf{d}=\left(1,-\frac{1}{2^3},\frac{1}{3^3},-\frac{1}{4^3},\frac{1}{5^3}\right)=(1,4,9,6,3)\in\mathbb{F}_{11}^5. \tag{3.4}$$

This follows from the diagonal action in Lemma 3.1 and $\bar{a}=-1$. The exponent three is essential: the odd elliptic point has $v_K(y')=1$, unlike the even cases.

3.3. **The final odd directions.** Here is an explicit computation of the Kummer image of (3.4). Choose smooth points with residual X-coordinates $r_i=0,2,3,4,8$ and square roots $y_i\in k$. At zero take $y_i=-b$. The tangent matrix of the five Abel maps is

$$M_{ij}=\frac{r_i^{j-1}}{2\bar{y}_i}.$$

It is invertible. Solve $\mathbf{u}M=\mathbf{d}$ over $\mathbb{F}_{11}$ and lift the u_i to integers. Let

$$P_i^\pm=(r_i\pm\varpi u_i,y_i^\pm),\qquad y_i^\pm\equiv y_i \pmod{\varpi}.$$

The anti-invariant formal divisor $6\sum_i(P_i^+-P_i^-)$ has leading tangent vector $\mathbf{d}$, because $12\equiv 1 \bmod 11$. Its Kummer image is therefore

$$d_v=6\sum_i\big(\mu_v(y_i^+-b)-\mu_v(y_i^--b)\big). \tag{3.5}$$

By (3.1), this is the image of every anti-invariant formal point with that leading tangent vector. The calculations check $d_v\neq 0$ and membership in the full local image.

Let $\mathcal{S}_R^+$ be the restriction of $\mathcal{S}^+$ to the two ramified places, and let

$$\mathcal{D} = \langle (d_{v_1}, 0), (0, d_{v_2}) \rangle.$$

The exact finite-field certificates give

n_0	$\dim \mathcal{S}^+$	localization kernel	$\dim \mathcal{D}$	$\dim(\mathcal{S}_R^+ \cap \mathcal{D})$
-1	1	0	2	0
1	1	0	2	0
3	1	0	2	0

We retain the global and local matrices, as well as the exact global representatives. Separate finite-field linear algebra reconstructs the Selmer upper bounds and verifies these intersections.

Proposition 3.3. *If $n_0 \in \{-1, 1, 3\}$, $n \equiv n_0 \bmod 4356$, and (2.3) holds, then $n = n_0$.*

Proof. Suppose otherwise and form D as above. Its local Kummer images at the two ramified places vanish, since its formal depth is at least three. By (2.7), its global Kummer class is zero. Hence there is a global λ-division. At each ramified place the division is unique, because the twisted λ-torsion is zero. It is therefore exactly the formal division of Lemma 3.1. Repeat this argument while the depth is at least three. After the r divisions in (3.3), one obtains a global point whose pair of local Kummer images is in $\mathcal{S}_R^+$. Equations (3.4)–(3.5) show that this pair is a nonzero element of $\mathcal{D}$, contradicting the computed zero intersection. □

3.4. **The even classes.** We now let $n_0 \in \{-2, 0, 2, 4\}$. The curve (2.5) is already defined over F, and $J[\lambda](F) = \langle T \rangle$ has order 11. At the three primes above 11 we have $e = 1$, $\bar{a} = 8$, and $\overline{Y(P_0)} \neq 0$. Thus

$$\lambda : H_m \xrightarrow{\sim} H_{m+1} \qquad (m \geq 1) \tag{3.6}$$

by Lemma 3.1. At each such place write $k = F_v$. We use the Abel map $i : C \to J$, $i(P) = [P - P_0]$.

Let $\mathcal{S}$ be the localization of the λ-Selmer group at the three primes over 11. Define

$$U = \prod_{v|11} \mu_v(H_1), \quad T_{\mathrm{loc}} = \prod_{v|11} \langle \mu_v(T) \rangle, \quad T_{\mathrm{glob}} = \langle (\mu_v(T))_v \rangle.$$

The exact Selmer computations give

n_0	-2	0	2	4
$\dim \mathrm{Sel}^\lambda(J/F)$	5	5	4	4
$\dim \ker(\mathrm{loc}_{11})$	0	0	0	0
$\dim U$	3	3	3	3
$\dim T_{\mathrm{loc}}$	3	3	3	3

and

$$U \cap T_{\mathrm{loc}} = 0, \qquad \mathcal{S} \cap (U + T_{\mathrm{loc}}) = T_{\mathrm{glob}}. \tag{3.7}$$

Every local torsion image is nonzero. The class-group vanishing (2.6) also applies here. The split descent algebra is $F \times F$, whose global S-class space has dimension 38. The calculation certifies 38 candidate classes by exact ideal membership and independent joint local images. It then imposes the norm and local conditions, including the condition at 2. This certifies the global space without a GRH assumption. The identities (3.7) are verified from the saved localization matrices by exact linear algebra.

Proposition 3.4. *If $n_0 \in \{-2, 0, 2, 4\}$, $n \equiv n_0 \bmod 4356$, and (2.3) holds, then $n = n_0$.*

We first describe the local formal image used in (3.7). Here $\pi = 1 - \zeta_{11}$ is a uniformizer and $11/\pi^{10} \equiv -1 \pmod{\pi}$. To compute $\mu_v(H_1)$, choose smooth sections with distinct residual X-coordinates $0, 2, 3, 4, 8$. At $X = 0$ choose $Y = -b$. The sum of the five Abel maps is a system of formal coordinates, since its differential is a Vandermonde matrix with unit determinant. For a displacement $X \mapsto X + \pi u$, direct expansion gives

$$\frac{Y_{\mathrm{new}} - b}{Y_{\mathrm{old}} - b} \equiv \begin{cases} 1 + \pi^{11}\bar{u}/(4\bar{b}^2) & \bar{X} = 0, \\ 1 & \bar{X} \in \{2, 3, 4, 8\} \end{cases} \pmod{\pi^{12}}.$$

Indeed, at a nonzero residue the two possible terms of valuation 11 in $(X + \pi u)^{11} - X^{11}$ cancel: $(11/\pi^{10})\bar{X}^{10}\bar{u} + \bar{u}^{11} = 0$. All elements of $1 + \pi^{12}\mathcal{O}_k$ are eleventh powers, by logarithm and exponential on $1 + \pi^2\mathcal{O}_k$. The first line generates the nonzero unramified Kummer line. Its nonvanishing is also verified by the local Kummer maps in the computation. Thus $\dim \mu_v(H_1) = 1$.

Let

$$\ell_0(X) = \frac{(X-2)(X-3)(X-4)(X-8)}{(-2)(-3)(-4)(-8)} = 1 + 2X + 2X^2 + X^3 + 9X^4 \in \mathbb{F}_{11}[X].$$

This is the linear functional extracting the displacement at the section with residual coordinate zero, expressed in the differential basis. For a point P near P_0, suppose $v_k(X(P) - X(P_0)) = d \geq 2$. Then $i(P) \in H_d \setminus H_{d+1}$. Put $Q = \lambda^{-(d-1)} i(P) \in H_1$, using the unique formal divisions in (3.6). Up to a nonzero scalar, its Kummer image is

$$h_{d-1}(8), \qquad h_r(x) = \sum_{j=1}^{5} \ell_{0,j-1}\, x^{j-1} j^{-r}, \tag{3.8}$$

where $\ell_{0,j-1}$ is the coefficient of X^{j-1} in ℓ_0. This follows by applying the diagonal leading action of successive formal divisions to the Abel tangent vector $(1, 8, 8^2, 8^3, 8^4)/(2\overline{Y(P_0)})$. All nonlinear terms disappear on the associated graded groups. In particular,

$$r \equiv 1 \pmod{10} \quad \Longrightarrow \quad h_r(8) = h_1(8) = 5 \neq 0. \tag{3.9}$$

Normalize $v_K(u) = 1$ as above, and use the same integral minimal model. Its discriminant has valuation three. The point $2G$ is formal, with parameter $-x'/y' = 4u$. If $n \neq n_0$, let $k_0 = v_{11}(n - n_0) \geq 2$. Since $n - n_0$ is even, the elliptic formal logarithm, which is an isometry here, gives

$$v_K(t((n - n_0)G)) = 1 + 5k_0.$$

For $q_0 = n_0 G - 3s$, the exact computations give

$$v_K(x(q_0) - 4) = 2, \quad v_K(y(q_0) + 1/2) = 3, \quad v_K(\delta_{n_0}) = 2.$$

Thus q_0 is a smooth point on the special fibre of the minimal model, with unit y', and translation by the formal point gives

$$v_K(x(nG - 3s) - x(q_0)) = 3 + 5k_0.$$

Equation (2.3) now yields

$$v_K(w^{11} - 1) = 1 + 5k_0, \qquad v_K(w - 1) = 5k_0 - 4.$$

For the second equality, first reduce modulo u to get $\bar{w} = 1$. In the binomial expansion the term $11(w - 1)$ then has strictly smaller valuation than all other nonzero terms, since $v_K(11) = 5$. At every $v \mid 11$ of F we obtain

$$d = v_k(X(P) - X(P_0)) = 10k_0 - 8 \geq 12, \qquad d - 1 \equiv 1 \pmod{10}. \tag{3.10}$$

We choose the global square root so that P is near P_0 at all three places. Put

$$R = (n/2)G + 4s, \qquad R_0 = (n_0/2)G + 4s,$$

so $2R = nG - 3s$. Then

$$v(R) = \frac{(x(R) - \theta)^2 - (3\theta^2 - 2\theta - 7)}{2y(R) + 1} \tag{3.11}$$

satisfies $v(R)^2 = x(2R) - \theta$. At R_0 this equals the chosen known square root exactly. Since $4 \mid M$, the difference $R - R_0 = ((n - n_0)/2)G$ is formal. Its parameter has valuation at least 22 over each F_v. All R_0 have integral minimal coordinates. The computed valuations are

$$4 \le v_k(x(R_0) - \theta) \le 6, \quad v_k(2y(R_0) + 1) \le 8, \quad v_k(v(R_0)) = 2.$$

Translation in the minimal coordinates changes the original x and y coordinates by quantities of valuations at least 26 and 28. Substituting in (3.11) shows $v(R)/v(R_0) \equiv 1 \pmod{\pi^{20}}$. This fixes the required signs simultaneously. Since $2Y(P_0)$ is a unit, P is in its smooth residue disc, and (3.10) applies.

Lemma 3.5. *Under the Selmer injectivity and intersection conditions in (3.7), no global point $D \in J(F)$ can satisfy $D \in \prod_{v|11} H_{d,v}$, with $d \ge 2$, and have a nonzero Kummer image after its $d - 1$ successive formal λ-divisions.*

Proof. Write Q_j for the formal quotient $\lambda^{-j}D$, for $0 \le j \le d-1$. Initially $Q_0 = D$ is global, and its local Kummer image is zero because $H_d \subset \lambda H_1$. Injectivity of Selmer localization implies that its global Kummer class is zero. Hence a global R with $\lambda R = Q_0$ exists. Locally, $R = Q_1 + t_v T$, for scalars $t_v \in \mathbb{F}_{11}$. Its local Kummer image lies both in $\mathcal{S}$ and in $U + T_{\mathrm{loc}}$. By (3.7), the formal contribution is zero and all t_v coincide. Subtracting this global multiple of T shows that Q_1 is global and has zero global Kummer class, the latter again by localization injectivity. Repeating the argument shows that the local Kummer image of Q_{d-1} is zero, as asserted. □

Proof of Proposition 3.4. If $n \ne n_0$, the construction above gives $D = i(P) \in J(F)$ with depth $d = 10k_0 - 8$ at every place above 11. Equations (3.8)–(3.9) show that its final formal quotient has nonzero Kummer image at every such place. This contradicts Lemma 3.5. □

This is the division-image argument of Selmer-group Chabauty, applied to the isogeny λ. See [6, Theorem 2.1 and Remark 2.2].

Proof of Theorem 1.1. The six-prime sieve puts the image of every rational point into one of the seven states (2.2). The cusp identity gives (2.3). Propositions 3.3 and 3.4 then give $n = n_0$. The seven base points are listed in Table 1, and each has its unique rational CM lift. There are no rational cusps, since their images would be rational cusps on E, whose cusp divisor has degree five and is irreducible. This proves the assertion. □

4. Computational data

We use Magma V2.30-1 [1] for the algebraic computations and Python for the finite sieve. The ancillary data contain the exact inputs, completed logs and the finite-field matrices used in the proof. We describe the necessary calculations and the supplementary checks separately.

4.1. **The sieve and the auxiliary curves.** We construct the six-prime tables directly from the normalized functions f_m. The program `six_prime_sieve.py` enumerates the possible classes and congruences. It retains every cusp residue, so no additional criterion at a cusp is needed. The exact computation of the known CM classes gives the seven states in (2.2).

The files `verify_descent.m` and `verify_genus5.m` check the normalized function identity, the number fields and the seven genus-five equations. The valuation and square-root identities used in the local argument are checked by `verify_even_geometry.m`, `verify_odd_geometry.m` and `verify_odd_depth.m`.

4.2. **The Selmer calculations.** For each odd class, we give fifteen exact elements of B and their joint local images. We check that the valuations outside S are divisible by 11. Their independent local images, together with the proved global dimension, certify the global basis. We generate each of the three local Kummer images using actual local divisors until its dimension is five. The resulting matrices determine the Selmer upper bound and verify (2.7) and the zero intersections with the two formal direction lines.

For each even class, the data give the Selmer localization, the three formal lines and the three local torsion lines. Exact linear algebra verifies (3.7). The global basis certificate and the local conditions at 2 and 11 are included in the reproduction scripts.

GRH is used to find candidate generators, but each candidate is subsequently checked by ideal factorization and local independence. The required dimensions follow from (2.6). Thus the global bases and the resulting Selmer bounds are certified without GRH.

The odd calculation is recorded at precisions 80 and 160. We accept a local Kummer evaluation only when its relative precision is at least $2v(11)+2$, which exceeds the eleventh-power detection threshold. The finite-field matrix audits do not depend on a further local approximation. The formal-group lemmas in Section 3 give the mathematical argument relating these data to rational points.

4.3. **Supplementary checks and reproduction.** The cases $n = n_0 + 4356$ and $n = n_0 + 11 \cdot 4356$ give additional checks of the odd depth and direction formulas at both ramified places. They are checked by `verify_local_depth_examples.m`. These examples are not used in place of the argument for arbitrary $v_{11}(n - n_0)$ in Section 3.

The command `audit_saved_certificates.py` repeats the finite sieve, compares its surviving classes with the exact CM classes and verifies all saved odd and even matrix identities. It does not repeat the number-field calculations or generate local points. The ancillary instructions give separate commands for the full descents and the exact geometry checks, together with a manifest of SHA-256 hashes.

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